ACADEMIC MINI PROJECT 2026 • COMPUTER SCIENCE & ENGINEERING

Editorial Workflow & Content Studio

A Browser-Native Content Lifecycle & Kanban Publishing Platform

A clean, responsive, and lightweight editorial content studio designed to manage academic and technical publishing pipelines. Features interactive Kanban transitions, real-time dashboard metrics, and client-side persistence—operating purely in browser memory with zero backend dependencies.

Technology: Semantic HTML5 • Modern CSS3 • Bootstrap 5.3 • Vanilla JavaScript (ES6)

PROJECT TEAM & COLLABORATION

Project Team Details

B.Tech Computer Science & Engineering • Academic Project Contributors

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Dev Shukla

Frontend Lead

Bootstrap 5.3 UI architecture, responsive component styling, layout responsiveness.

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Ayush Gaur

Editorial Logic

Kanban workflow modeling, peer review lifecycle validation, and test flows.

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Devansh Goyal

JavaScript Architect

Vanilla JS state controller, DOM manipulation, LocalStorage synchronization engine.

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Anushka Chauhan

UI/UX & QA Specialist

Editorial checklist design, visual accessibility, cross-browser validation.

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Ashutosh Gupta

System Analyst & Docs

Requirement specifications, system flow mapping, academic documentation.

BACKGROUND & ANALYSIS

Problem Statement & Motivation

Understanding the bottlenecks in traditional collegiate publishing systems

MANUAL & CHAOTIC PIPELINES

Unstructured Submissions

- Articles submitted through emails, messaging chats, or raw documents lack revision control.
- Authors have zero visibility into whether submissions are in draft, under review, or approved.
- Editorial status is often maintained in scattered notes, causing lost drafts and duplicate work.

NO FORMAL APPROVAL GATE

Lack of Review Workflow

- Traditional blogs post directly without a structured peer review phase.
- Absence of role-specific checkpoints where editors can evaluate academic quality and formatting.
- No dedicated workflow board to inspect work column-by-column across department releases.

CUMBERSOME INFRASTRUCTURE

Heavy Backend Overhead

- Enterprise CMS solutions (WordPress, Drupal) require relational DBs, server hosting, and security patches.
- Overkill for academic mini projects, college magazines, or local campus publications.
- Need for a client-side, zero-install, zero-cost architecture running instantly in standard browsers.

SYSTEM OBJECTIVES

Proposed Solution & Key Goals

Delivering a lightweight, robust client-side editorial publishing solution

The Solution Concept

The Editorial Workflow & Content Studio is an integrated web-based editorial workbench. It gives student editorial teams complete control over content staging, peer review, and final release through a sleek, modern visual interface inspired by enterprise CMS design.

Core Design Principles:

✓ Zero-Latency Local Operation: Operates in browser memory & local storage.

✓ Strict Publishing Governance: Clear transitions from Draft to Published.

✓ Accessibility & Clarity: Professional high-contrast light typography.

✓ Modular Code Structure: Clean separation of concerns across pages.

1. Visual Kanban Governance

Column-by-column visual tracking of story lifecycle with one-click transitions.

2. Real-Time Dashboard KPI

Automatic aggregation of total volume, active reviews, and completion rates.

3. Frictionless Article Editor

Form for title, author, category, abstract, and monospace content editing.

4. Cross-Page Data Sync

Browser storage keeping all views instantly synchronized without page drift.

TECHNICAL BLUEPRINT

System Architecture & Editorial Pipeline

Three-tier client architecture and sequential lifecycle state machine

Three-Tier Client-Side Architectural Model

Presentation Layer

HTML5 & Bootstrap 5.3 CDN

Responsive views for Dashboard, Kanban Board, Editor & Live Landing.

Controller & Logic

Vanilla JavaScript Engine

Data parsing, dynamic DOM injection, status transitions & URL query routing.

Persistence Layer

Browser LocalStorage API

JSON-serialized store (editorial_articles) preserving state across page reloads.

Stage 01: Draft Creation

Authoring & Initial Input

- Contributor fills story metadata in editor.html
- Form assigns unique incremental integer ID
- Initial status tagged as 'Draft'
- Story appears in Drafts column

Stage 02: Peer Review

Evaluation & Editorial Gate

- One-click promotion: 'Review →'
- Article locked in review queue for senior inspection
- Editorial checklist verified by lead editor
- Feedback incorporated or approved

Stage 03: Live Release

Approved & Public Stream

- Final approval triggers 'Publish →'
- Article displays in Published column and Landing Table
- Dashboard KPI reflects updated completion rate
- Full view access enabled

SYSTEM CAPABILITIES

Core Functional Modules

Interactive breakdown of the studio's primary application interfaces

DASHBOARD.HTML

Executive Dashboard

High-Level Metrics

Calculates total articles, active review tickets, and live published stories in real time.

Dynamic Progress Bars

Calculates percentage metrics reflecting editorial completion rates.

Pipeline Data Table

Detailed list view showing article title, subject badge, author avatar, stage, and actions.

Direct Row Controls

Instant access to edit story details or delete outdated drafts with confirmation prompts.

BOARD.HTML

Editorial Kanban Board

Three-Column Workflow

Visual columns for Drafts, In Peer Review, and Published Releases.

Topic Category Filtering

One-click pill filtering across Web Dev, Database, and Networking stories.

Stage Transition Triggers

Direct interactive buttons to move cards forward (Review →, Publish →) or backward.

Story Count Badges

Dynamic header counts indicating volume of content in each phase.

EDITOR.HTML

Article Studio Form

Dual Operation Mode

Handles both new article creation and editing existing articles via URL query parsing (?id=X).

Structured Inputs

Clean fields for title, author, topic category, workflow stage, abstract, and content body.

Editorial Checklist

Built-in verification sidebar to ensure academic quality before submission.

Flexible Submission

Dual actions: 'Submit for Review' (promotes stage) or 'Save Draft' (preserves draft stage).

ENGINEERING FOUNDATION

Technical Stack & JavaScript Implementation

Direct application of classroom JavaScript concepts to solve publishing challenges

Technologies Utilized

HTML5 Semantic Layout: nav, main, table, form, and header tags for structured accessibility and screen readers.

Modern CSS3 System: Custom color tokens, flexbox alignment, card hover elevation, and sidebar mechanics.

Bootstrap 5.3.3 CDN: Responsive 12-column grid, responsive tables, badge components, and button utilities.

Bootstrap Icons 1.11: Vector SVG icon suite illustrating stage indicators, navigation, and user roles.

Git Version Control: Feature branching, iterative commits, and synchronized pull request workflow on GitHub.

Classroom JS Concepts Applied

Variables & Declarations: let and const scoping managing state collections, counters, and references.

Data Types & Objects: Structured article models { id, title, category, stage, author, abstract }.

Operators & Math: === strict comparison, ++ counters, and Math.round percentage progress calculations.

Control Flow: if / else if structures checking stage transitions, URL params, and category matches.

Loops & Iteration: Standard for loops traversing article arrays for dynamic dashboard & board rendering.

Array Operations: .push() for appending new stories, .splice() for atomic deletion, .length for bounds.

Functions & Modularity: Dedicated modular functions: getArticles, saveArticles, renderBoard, changeStage.

USER EXPERIENCE

UI/UX & Editorial Design System

Principles of human-centered academic design, clean ergonomics, and visual hierarchy

EXECUTIVE ACADEMIC PALETTE

High-Contrast Readability

- Light background canvas (#F8FAFC) minimizing cognitive fatigue.
- Deep slate typography (#0F172A) exceeding WCAG AA accessibility contrast.
- Monospaced editor textarea for precise, distraction-free drafting.

INSTANT COGNITIVE RECOGNITION

Visual Stage Encoding

- Color-coded badges: Gray for Draft, Amber for In Review, Green for Published.
- Dynamic progress bars showing completion percentage at a single glance.
- Standardized status indicators across all dashboard and board views.

UNIFIED SIDEBAR & BREADCRUMBS

Ergonomic Navigation

- Fixed 240px sidebar with dedicated icons for Home, Dashboard, Board, and Editor.
- Direct CTA buttons allowing one-click transition between writing and review.
- Responsive collapse for tablet and mobile viewport screens.

SAFETY & CONFIRMATION MECHANICS

Feedback & Error Prevention

- Mandatory title validation preventing accidental submission of empty drafts.
- Window confirmation alerts prior to irreversible story deletion.
- URL query param (?id=X) ensuring users never overwrite unrelated records.

OUTCOMES & ROADMAP

Conclusion & Future Enhancements

Summary of project milestones achieved and roadmap for future iterations

Key Project Achievements

✓ Functional Editorial Staging

Implemented full multi-page lifecycle tracking articles from draft to live release without server overhead.

✓ Real-Time Kanban Interaction

Built interactive column boards enabling editors to move, inspect, and approve stories with single clicks.

✓ Zero-Backend Persistence

Engineered a reliable browser storage mechanism that maintains full article state across page sessions.

✓ Classroom Concept Mastery

Demonstrated practical mastery of variables, loops, objects, arrays, and control flow in a realistic software product.

Future Enhancement Scope

✦ Role-Based Access Control (RBAC)

Integrating authentication distinguishing author submission rights from editor approval permissions.

✦ Cloud Synchronization (Firebase / Supabase)

Adding multi-device sync and concurrent real-time collaborative editing for distributed teams.

✦ Automated Article Exporting

Single-click export of approved publications into formatted academic PDFs and Markdown bundles.

✦ Editorial Analytics & Read Time

Algorithmic word count analysis, reading duration estimation, and author productivity telemetry.

EDITORIAL WORKFLOW & CONTENT STUDIO

Thank You!

Questions & Technical Discussion

We sincerely appreciate your time, attention, and valuable feedback on our academic mini project.

The complete codebase and interactive demo are live on GitHub.

Project Repository: https://github.com/Devansh7006/Editorial-Workflow-Content-Studio-

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